LOW-CODE EXPERIMENT

Report on Career Pathways and the Future of Low-Code Development

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Abstract

This report explores career pathways created by low-code technologies, the technical and professional skills they require, and the likely future of low-code development. It outlines key roles, learning routes, and the influence of artificial intelligence and governance. It also explains how low-code development can help an AI and machine learning student turn ideas into usable applications.

1. Introduction

Low-code development allows applications to be built using visual tools, forms, reports, workflows, and limited scripting. It reduces the amount of code needed, but still requires problem-solving, data modelling, process design, integration, testing, and security. These skills are valuable in organisations that want to build and improve applications efficiently.

2. Career Pathways

Role	Responsibility	Key Skills
Low-Code Developer	Builds applications using forms, reports, workflows and scripts.	Data modelling, Deluge or similar scripting, UI design and testing.
Citizen Developer	Creates simple tools to solve departmental problems.	Domain knowledge, basic logic and platform fundamentals.
Business Analyst	Maps business processes into automated workflows.	Requirement gathering, process modelling and communication.
Integration Developer	Connects low-code applications to other systems.	REST APIs, JSON, OAuth and middleware.
Solution Architect	Designs solutions combining low-code and traditional development.	System design, security, scalability and governance.
Platform Administrator	Manages users, roles, environments and compliance.	Role-based access control, audits and deployment management.
Pro-Code Extension Developer	Builds custom functions and services to extend a platform.	Java, Python or Node.js, plus serverless platforms.
AI and Automation Specialist	Adds AI, analytics and hyperautomation to applications.	ML fundamentals, analytics tools and prompt design.

3. Skills to Build

Technical skills: relational data modelling; workflow scripting such as Deluge; REST APIs and integration; security and role-based access; cloud and DevOps basics; AI and analytics fundamentals.

Professional skills: requirement analysis, problem-solving, communication with stakeholders, process thinking, documentation, project and time management, and continuous learning.

Technical knowledge helps when applications become complex, while professional skills ensure that the application meets users' real needs.

4. Certification and Learning Routes

- Follow vendor learning paths and certifications, including Zoho Creator resources.
- Explore transferable skills on Microsoft Power Platform, OutSystems, Mendix and Appian.
- Build a portfolio of working applications and document projects on GitHub.
- Participate in hackathons, internships and developer communities.

5. Future of Low-Code Development

• **AI-native building:** users will increasingly describe an application in plain language and refine it conversationally.
• **Low-code and pro-code convergence:** developers will use visual tools for speed and write custom code where necessary.
• **Stronger governance:** organisations will manage access, security, compliance and deployment more carefully.
• **Wider adoption:** reusable templates will support fields such as healthcare, hospitality and education.
• **Evolving roles:** as routine form and report creation becomes more automated, architecture, integration, data and AI skills will become more valuable.

6. Relevance to an AI and ML Student

Low-code platforms can help AI and ML students turn trained models into usable products. A model can be made available through an API and connected to a low-code interface that provides forms, user management and dashboards. Combining machine-learning knowledge with low-code delivery and API integration can create practical, end-to-end solutions.

Example from this lab: the Patient Appointment Booking System uses forms to collect appointment details, workflows to validate and process submissions, permissions to control access, and a dashboard to summarise registrations. It demonstrates how a low-code application can solve a real-world workflow problem.

7. Conclusion

Low-code development offers career opportunities that combine technical ability with business understanding. As AI makes application building easier, long-term value will increasingly depend on sound data models, integrations, security and intelligent automation. Students can prepare by learning platform fundamentals and building a documented portfolio.

8. Summary

The report describes eight low-code career roles, their key skills, learning routes and future trends. It highlights AI-assisted building, the combination of low-code and traditional programming, stronger governance, and the growing importance of architecture, integration, data and AI.

9. Learning Outcome

After completing this experiment, the learner can identify career options in low-code development, list the skills and learning routes needed for those roles, and explain how the field may evolve.

10. Presentation Outline

1. Introduction to low-code development
2. Reasons for its growth
3. Career roles and responsibilities
4. Skills and certification roadmap
5. Future trends: AI, governance and hybrid development
6. Conclusion and questions